

Ch-4 $\rightarrow$ Number Theory & Cryptography

4.1

① Defⁿ 1: if $a, b \in \mathbb{Z}$, $a \neq 0$ then $c \in \mathbb{Z}$ such that $b = ac$
"a divides b" $\rightarrow b/a$ or $a|b$

eg: $3/7$ $12/24$

② Defⁿ 2:

Divisor (d)	dividend (a)	quotient (q)
remainders (r)		
$q = a \text{ div } d$ $= d a$ $= \lfloor a/d \rfloor$	$(r) = a \bmod d$	

Theorem-2

$$a = d \cdot q + \boxed{r} \rightarrow 0 \leq r < d$$

$$\boxed{b} = \underset{\uparrow}{a} \underset{\uparrow}{c} + \boxed{0}$$

⑧ Defⁿ 3: if $a, b \in \mathbb{Z}$, $m \in \mathbb{Z}^+$ then
 $a \equiv b \pmod{m}$ if $m \mid (a-b)$ $\frac{a-b}{m} \rightarrow r=0$
 ex) $a \not\equiv b \pmod{m}$

eg $\equiv 101 \equiv 2 \pmod{11}$
 $\downarrow \quad \downarrow \quad \downarrow$
 $a \quad b \quad m$

$11 \mid (101 - 2)$
 $\hookrightarrow 11 \mid 99 \checkmark$

$\left\{ \begin{array}{l} 100 \equiv 2 \pmod{11} \quad \times \\ 11 \nmid 98 \end{array} \right.$

⑧ Theorem = 3: if $a, b \in \mathbb{Z}$ and $m \in \mathbb{Z}^+$ then

$a \equiv b \pmod{m}$ iff $a \bmod m = b \bmod m$

$\hookrightarrow a, b \rightarrow$ have same remainder upon division by m

$m \quad 11 \mid 101 \mid 9$
 $\downarrow \quad 99 \mid$
 $\textcircled{2}$

$m \quad 11 \mid 2 \mid 0$
 $\downarrow \quad 0 \mid$
 $\textcircled{2}$

④ Theorem-4: let $m \in \mathbb{Z}^+$ $a \equiv b \pmod{m}$ is valid if $k \in \mathbb{Z}$

such that $a = b + km$

Derivation	
$a \equiv b \pmod{m}$	
$\rightarrow m \mid (a - b)$	
$\rightarrow (a - b) = km$	$b = ac$
$\rightarrow a = b + km$	

⑤ Theorem-5: let $m \in \mathbb{Z}^+$

then

$\rightarrow a \equiv b \pmod{m}$ (we prove)

$\rightarrow c \equiv d \pmod{m}$ (we prove)

$\rightarrow$ ① $(a + c) \equiv (b + d) \pmod{m}$ (we prove)

$\rightarrow$ ② $ac \equiv bd \pmod{m}$ (we prove)

⑧ Corollary-2 Let $m \in \mathbb{Z}^+$ and $a, b \in \mathbb{Z}$ then

① $(a+b) \pmod m = ((a \pmod m) + (b \pmod m)) \pmod m$

② $(ab) \pmod m = ((a \pmod m)(b \pmod m)) \pmod m$

⑧ Arithmetic modulo m $\begin{matrix} \xrightarrow{+} \\ \xrightarrow{-} \end{matrix} \begin{matrix} \boxed{0} \\ \boxed{1} \end{matrix} \xrightarrow{+}$ $\mathbb{Z}_m = \{ \quad \} \in \mathbb{Z}^+$

Let $\mathbb{Z}_m \in \mathbb{Z}^+$, such that $\mathbb{Z} \in \mathbb{Z}_m \pmod m$

$\rightarrow$ Addition $\rightarrow a +_m b = (a+b) \pmod m$

$\rightarrow$ Multiplication $\rightarrow a \cdot_m b = (ab) \pmod m$

eg: $7 +_{11} 9 = (7+9) \pmod{11} = 16 \pmod{11} = 5$
 $7 \cdot_{11} 9 = 63 \pmod{11} = 8$

4.2

Self study

- (5) inter conversions {
- 1. Binary expansion
 - 2. Octal u
 - 3. Hex n
 - 4. Base Conversion